01 LINE DRAWING

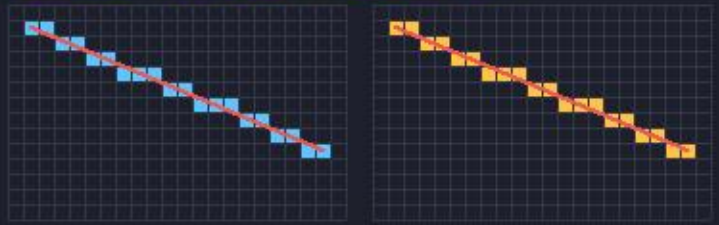
DDA + BRESENHAM: WIREFRAME BLOCKS, WORLD GRID, SELECTION

DDA VS BRESENHAM (ALL OCTANTS)



DDA BRESENHAM

PIXELS CHOSEN (ZOOM X8)



DDA (FLOAT) BRESENHAM (INT)

RED = IDEAL LINE (1,1)-(20,9)

PIXELS THAT DIFFER: 0

BRESENHAM: INTEGER MATHS ONLY

VOXEL WIREFRAME + WORLD GRID



DASHED = HIDDEN EDGES

LINE STYLES IN THE UI

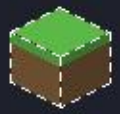
1 PX

3 PX

5 PX

DASH 10/5

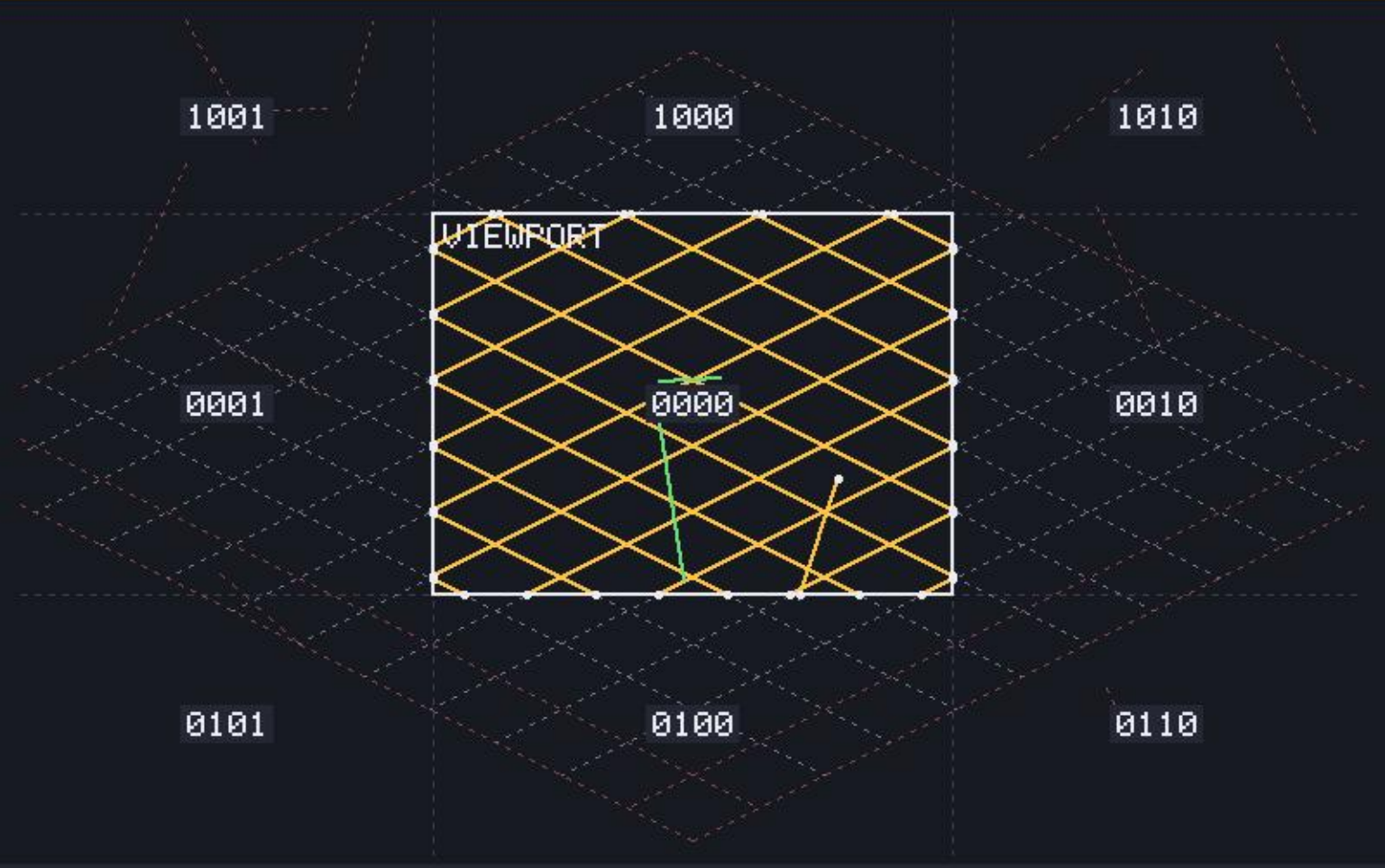
DASH 3/3



SELECTED BLOCK OUTLINE ->

02 LINE CLIPPING

COHEN-SUTHERLAND: CLIP THE WORLD GRID TO THE VIEWPORT



INSIDE: 2 CLIPPED: 21 REJECTED: 17 OUTCODE = TBRL

03 CIRCLES AND ELLIPSES

MIDPOINT ALGORITHMS: SUN, CLOUDS, CROSSHAIR, CHARACTER

CLOUDS = FILLED ELLIPSES

SUN = CIRCLE + RINGS



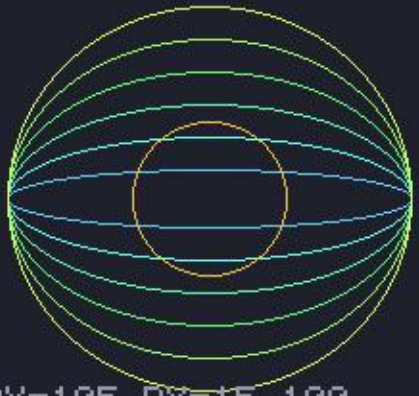
CROSSHAIR = CIRCLE + LINES

8-WAY SYMMETRY



8 PIXELS PER POINT

MIDPOINT ELLIPSE



RX=105 RY=15-100

CHARACTER + SHADOW

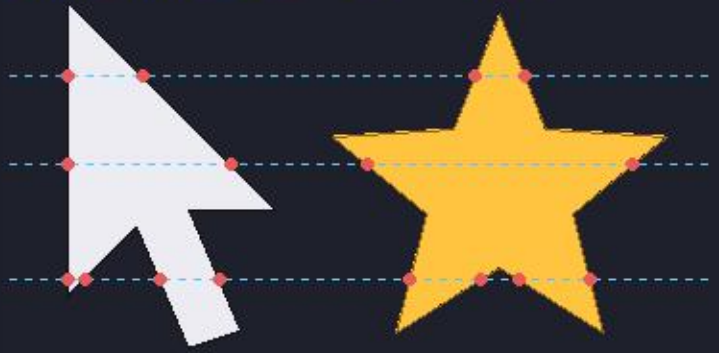


ONLY ROUND SHAPES

04 POLYGONS AND FILLING

SCANLINE (ET/AET), BOUNDARY FILL, FLOOD FILL, PROPERTIES

SCANLINE FILL (ET + AET)



RED = EDGE CROSSINGS, FILL PAIRS

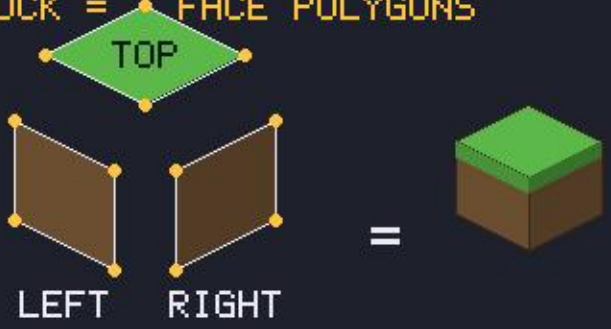
BOUNDARY FILL + FLOOD FILL



BOUNDARY
STOPS AT RED

FLOOD: GRASS->SNOW
SAME-COLOUR REGION

BLOCK = 7 FACE POLYGONS

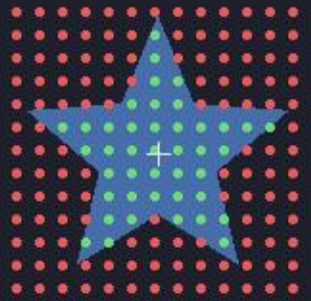


4 VERTICES PER FACE

POLYGON PROPERTIES



AREA 9987
CONVEX: YES

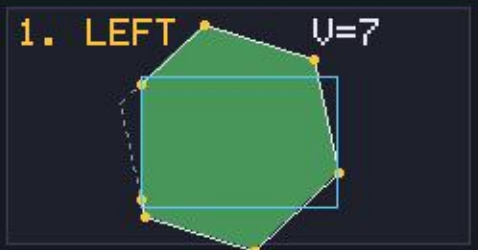


AREA 6771
CONVEX: NO

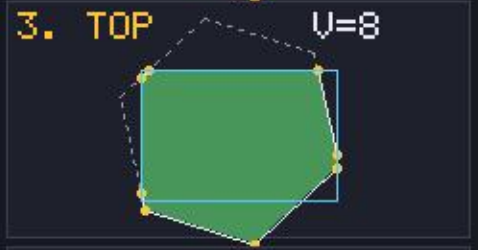
05 POLYGON CLIPPING

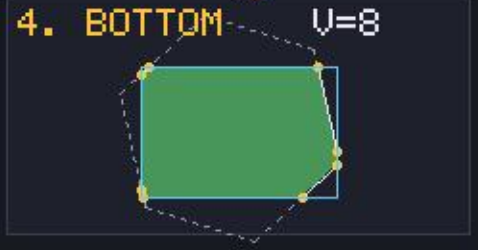
SUTHERLAND-HODGMAN: THE WORLD CLIPPED TO A VIEWPORT









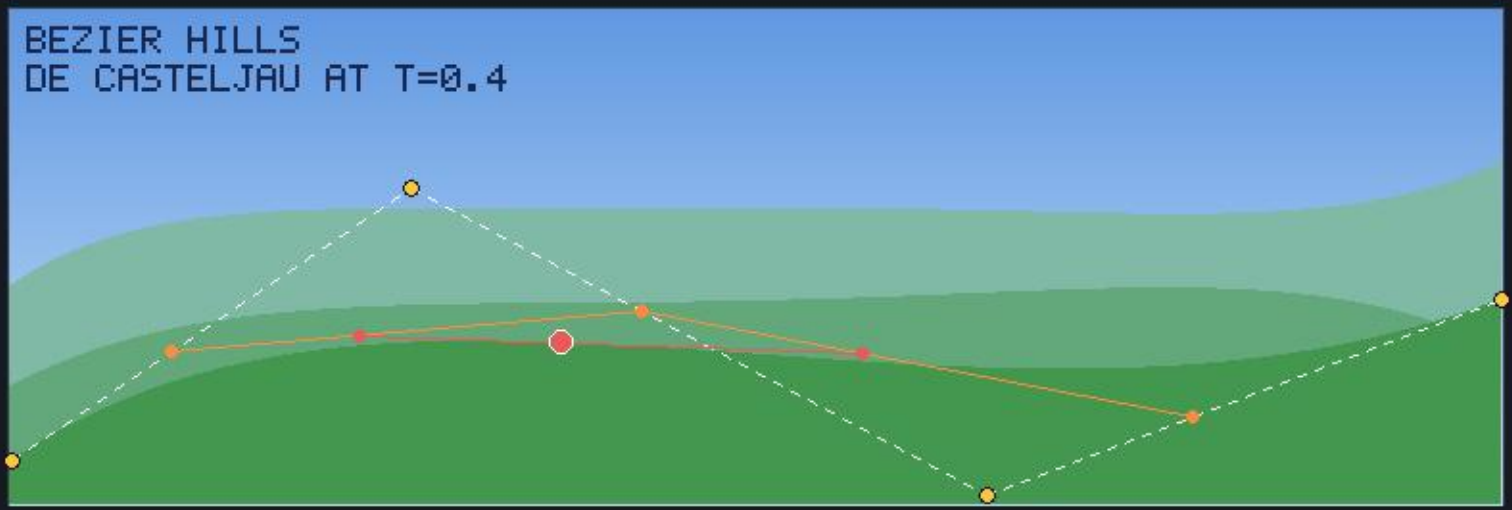


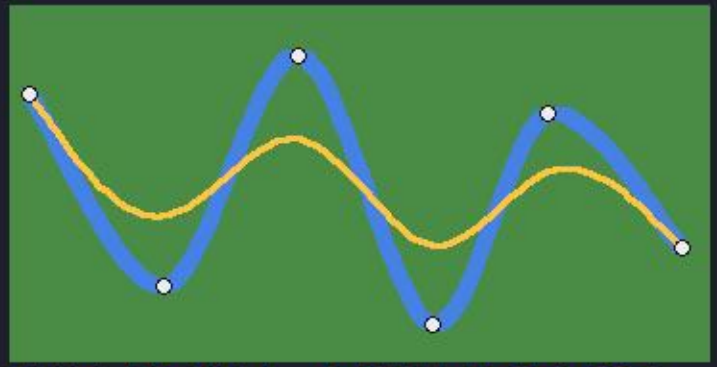
642 FACES CLIPPED BEFORE FILLING

FADED = OUTSIDE THE WINDOW

06 CURVES: BEZIER AND SPLINES

TERRAIN SILHOUETTES, A RIVER AND A CAMERA FLY-THROUGH







07 COLOUR THEORY

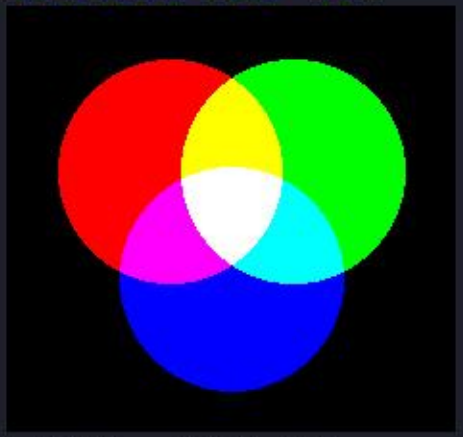
HSV WHEEL, ADDITIVE RGB, SUBTRACTIVE CMY, BLOCK SHADING

HSV COLOUR WHEEL



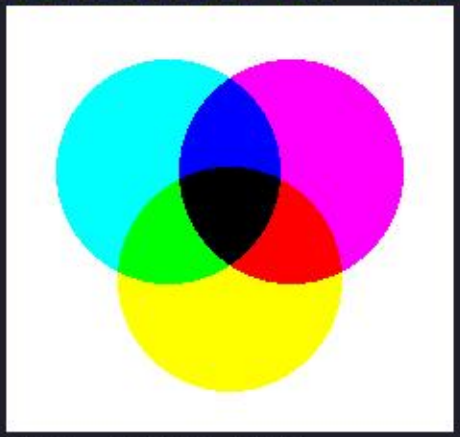
DOTS = BLOCK COLOURS

ADDITIVE RGB (OR)



LIGHT: SCREENS

SUBTRACTIVE CMY (AND)



INK: PRINTING

BLOCK PALETTE: FACE SHADES = HSV VALUE X 1.0 / 0.8 / 0.62



GRASS

H109 S60 V71
RGB 92,184,72

DIRT

H029 S55 V52
RGB 134,96,60

STONE

H240 S04 V52
RGB 128,128,134

SAND

H047 S34 V86
RGB 222,206,146

WATER

H216 S72 V84
RGB 58,120,216

WOOD

H027 S62 V36
RGB 92,60,34

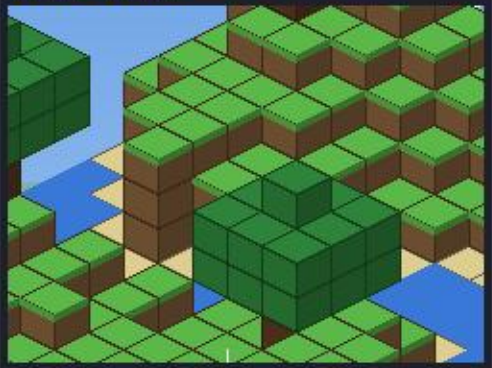
LEAVES

H128 S64 V51
RGB 46,132,58

08 COLOUR SPACE REDUCTION

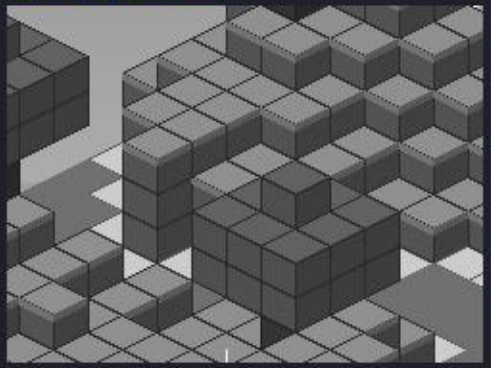
QUANTISATION AND DITHERING OF THE GAME SCENE

ORIGINAL



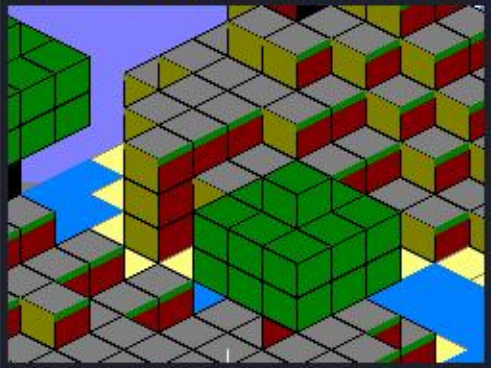
47 COLOURS

GREYSCALE



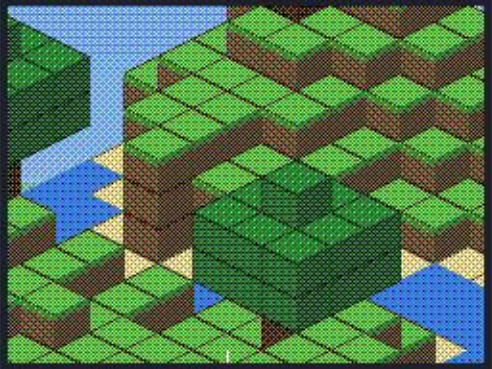
33 COLOURS

27 COLOURS: NEAREST



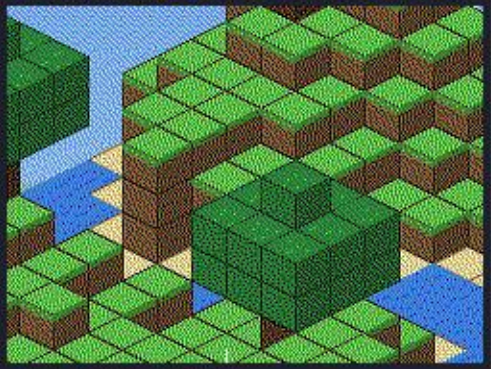
9 COLOURS

27 COLOURS: ORDERED



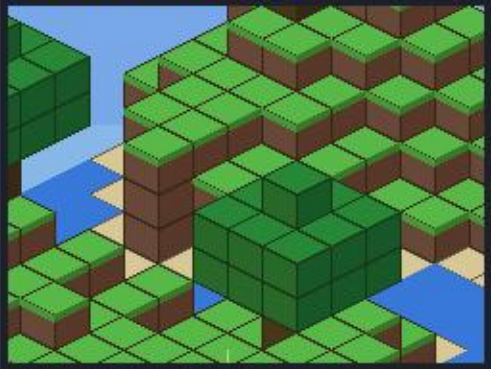
14 COLOURS

27: FLOYD-STEINBERG



22 COLOURS

16: POPULARITY

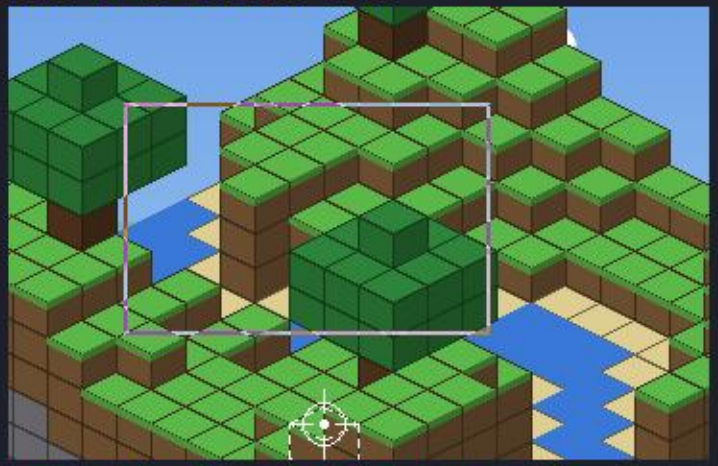


16 COLOURS

09 RASTER OPERATIONS

XOR CURSOR, BITBLT WITH COLOUR KEY, AND/OR SPRITE MASKING

XOR SELECTION BOX



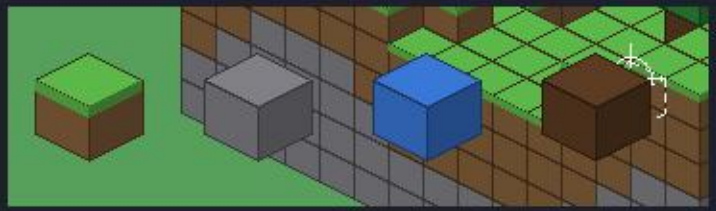
XOR TWICE = ORIGINAL: YES

BITBLT + COLOUR KEY

SPRITE SHEET (RAW)



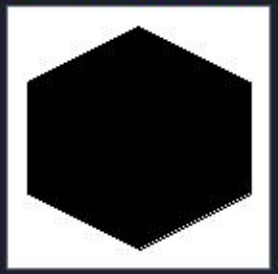
BLITTED, MAGENTA SKIPPED:



SPRITE MASKING: (BACKGROUND AND MASK) OR SPRITE



BACKGROUND



MASK



BG AND MASK



SPRITE



... OR SPRITE


VOXEL WORLD BUILDER
SELECTED: GRASS AT (9,3,11)
MINIMAP
GRASS

11 3D PREVIEW (TOWARDS CP2/CP3)

PERSPECTIVE, Z-BUFFER, FACE CULLING, DIFFUSE LIGHT

